Prescribing Diagnostic Imaging

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Learning Outcomes

1. Outline the effective use of Dento-maxillofacial Diagnostic imaging.
2. Describe the patient selection principles for screening radiography and other specific diagnostic tasks.

When to make a Radiograph???

- Performed only after a patient's thorough medical and dental history has been obtained and a clinical examination has been made.
- Guided by this examination and tailored specifically to the patient's needs and the diagnostic task.
- Potential diagnostic and treatment planning benefits outweigh the risks associated with radiation to the patient.

How to decide which Radiograph to make???

- Radiological Examination depends on
  1. The perceived nature or severity of an abnormality (including its size and accessibility).
  2. The ability of the imaging technique to accurately reveal the characteristic diagnostic features of the abnormality (sensitivity and specificity).
  3. The amount of image detail required (resolution).
  4. The radiation dose to the patient

INTRA-ORAL IMAGES

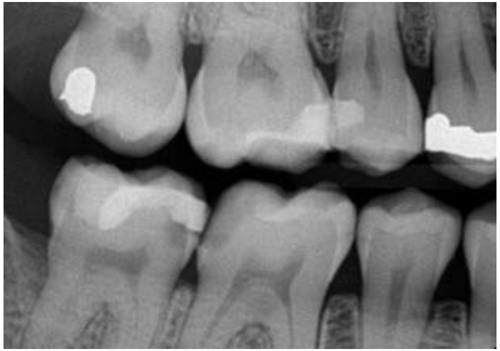
- Highest Spatial resolution.
- Tooth and supporting structures.

Intra-oral Periapical radiograph [IOPA]

- Image receptor is positioned parallel to the long axis of the teeth.
- Optimal to demonstrate the tooth root or roots, the supporting structures.
- A limitation of periapical imaging is geometric distortion.

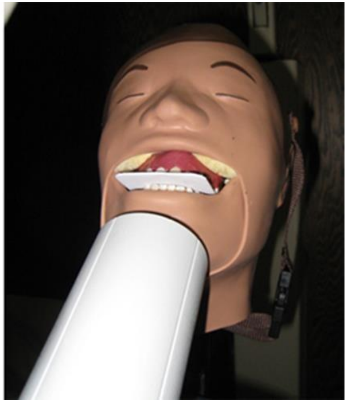
Bite-Wing

- receptor adjacent to the crowns of a sextant of the dentition and the crests of the alveolar processes.
- vertical bitewing
- optimal for revealing interproximal caries.



Occlusal X ray

- image receptor located between the occlusal surfaces of the teeth.
- supplement periapical and bitewing images.
- larger areas of the jaws can be imaged without compromising image resolution.



Extra-oral Images

- visualize a larger region of anatomy than intraoral images, but with much lower image resolution.
- image receptor located outside the mouth
- Includes
  1. Panoramic radiograph
  2. Advanced imaging techniques [CT, CBCT,MRI, Scintigraphy, PET Scan, Ultrasonography]

Panoramic imaging

Advantage

- broad view of the midface and jaws as well as adjacent structures

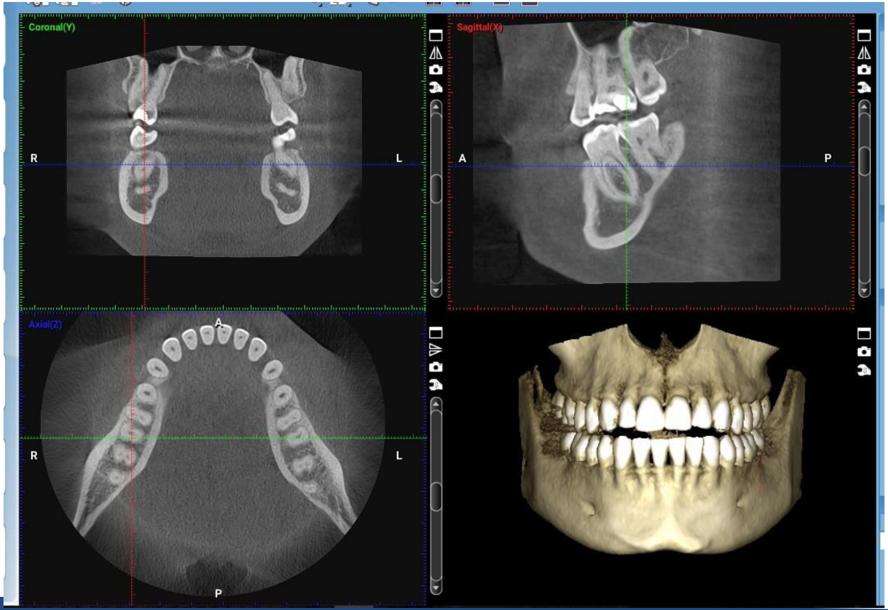
Disadvantage

- lower spatial resolution than intraoral images
- image distortion
- Superimposition
- patient movement during image acquisition

Advanced Imaging

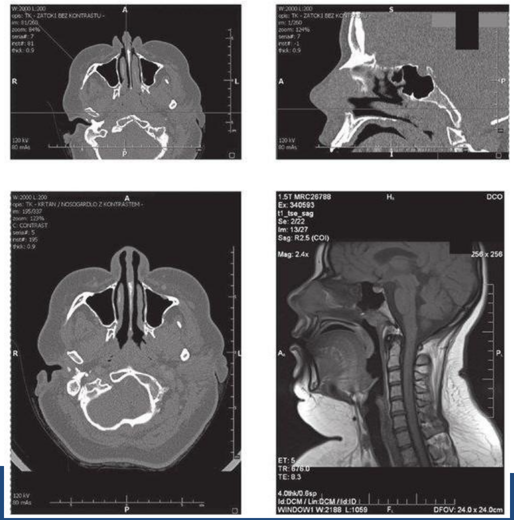
- CBCT, CT, MRI
- should be made if intraoral or panoramic imaging does not provide the necessary diagnostic and treatment planning information.
- Lower spatial resolution.
- Should be referred to radiologist for these procedures

CBCT



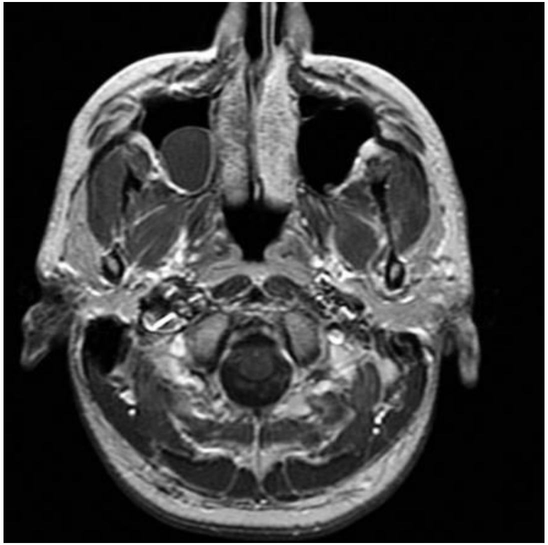
CT

- Cross sectional anatomy
- Bone pathology
- High radiation dose



MRI

- Non ionising
- Soft tissue`



Previous diagnostic images

- images may be helpful, regardless of when they were made.
- Comparison with current image.
- For all new patients, the dentist must review the patient's prior imaging history as part of the initial visit and attempt to obtain these images from the prior dental office.

Administrative images

- Insurance company: to approve a treatment.
- Typically do not benefit the patient.

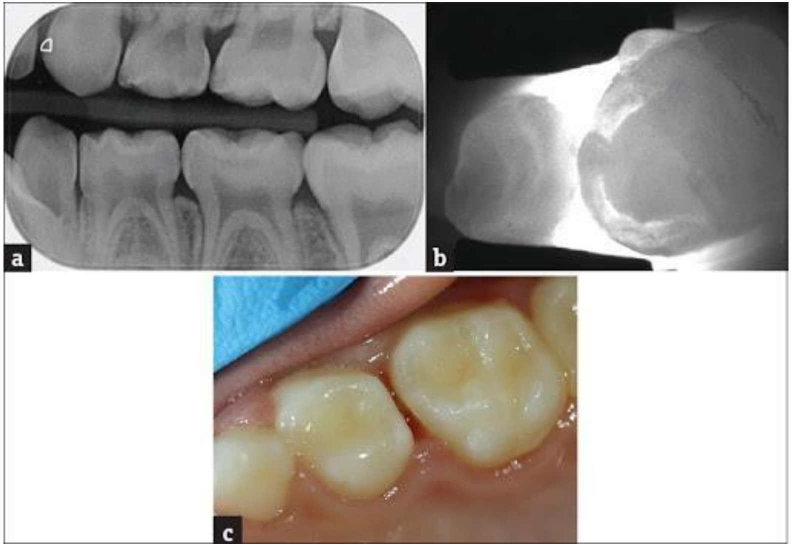
Guidelines for ordering dental radiographic examinations

- Where there is clinical evidence of an abnormality that cannot be fully assessed by physical examination alone.
- Where there is a high probability of disease that is not clinically evident. (Interproximal caries)

American Dental Association Selection Criteria for Prescribing Dental Radiographs

Positive Historical Findings	Positive Clinical Signs or Symptoms
1. Previous periodontal or endodontic treatment	1. Clinical evidence of periodontal disease
2. History of pain or trauma	2. Large or deep restorations
3. Familial history of dental anomalies	3. Deep carious lesions
4. Post-operative evaluation of healing	4. Malposed or clinically impacted teeth
5. Remineralization monitoring	5. Swelling
6. Presence of implants or evaluation for implant placement	6. Evidence of dental/facial trauma
	7. Mobility of teeth
	8. Sinus tract ("fistula")
	9. Clinically suspected sinus pathology
	10. Growth abnormalities
	11. Oral involvement in known or suspected systemic disease
	12. Positive neurologic findings in the head and neck
	13. Evidence of foreign objects
	14. Pain and/or dysfunction of the temporomandibular joint
	15. Facial asymmetry
	16. Abutment teeth for fixed or removable partial prosthesis
	17. Unexplained bleeding
	18. Unexplained sensitivity of teeth
	19. Unusual eruption, spacing or migration of teeth
	20. Unusual tooth morphology, calcification or color
	21. Unexplained absence of teeth
	22. Clinical tooth erosion
	23. Peri-implantitis

- Radiologic examination would yield clinically useful information.
- Consider each patient as an individual.
- It is inappropriate to irradiate the patient “just to see if there is something there.”
- Except: bitewing images for the detection of interproximal caries when no clinical signs exist of even early carious lesions.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Initial Visit

- For a child in the primary dentition who is cooperative and has closed posterior contacts, only bitewing images are required to detect interproximal caries.
- Additional imaging is recommended only when the medical or dental history or the clinical examination reveals cause.
- Children without evidence of abnormality and with open proximal contacts may not require a radiologic examination at this time.


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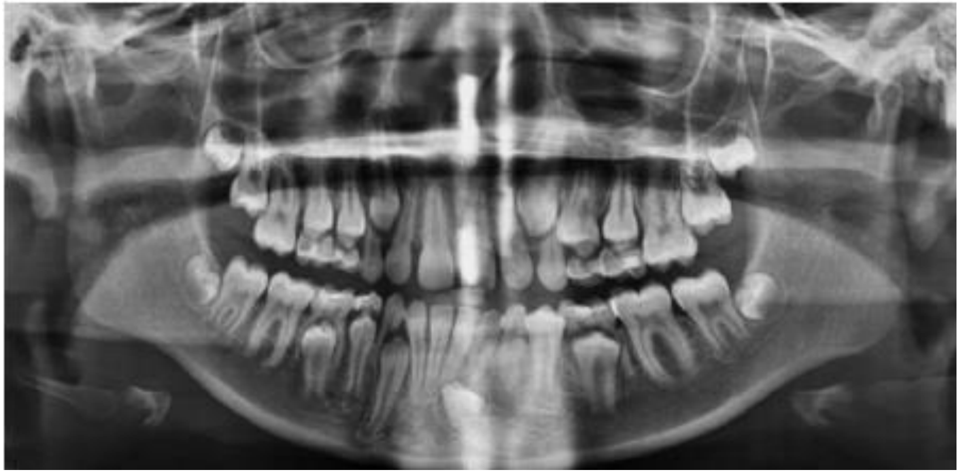
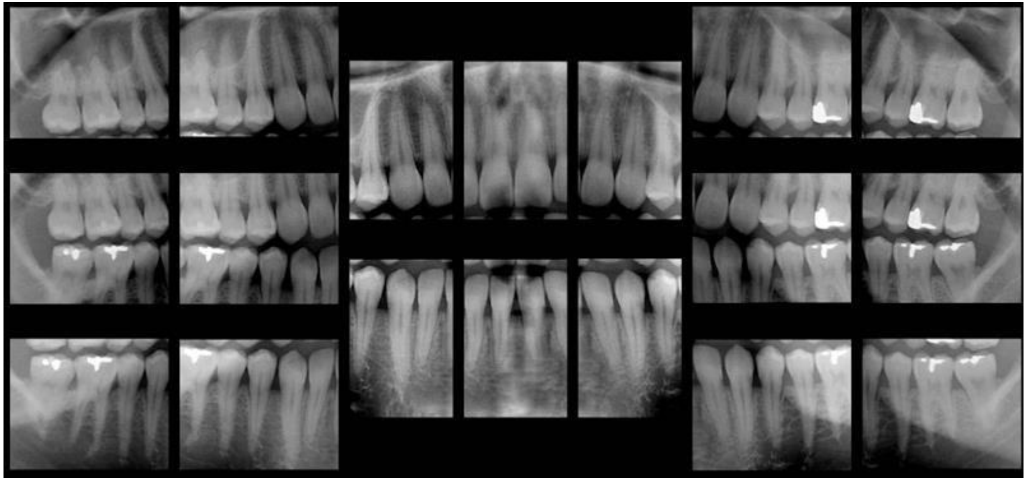
Child in the mixed dentition

- For a child in the mixed dentition following the eruption of a permanent first molar tooth, the guidelines recommend bitewing images to detect interproximal caries
- a panoramic image or selected periapical or occlusal images to detect abnormalities that may arise during growth and development.



Partially or fully dentate adult

- individualized radiologic examinations based on the patient's medical or dental history or if the clinical examination reveals cause for additional imaging.
- The presence of widespread disease (e.g., generalized caries or periodontal disease) may indicate a need for a full-mouth series of bitewing and periapical images.
- Alternatively, if disease is localized, a more limited examination consisting of a subset of these images may be completed.



Edentulous patient

- Individualized examination that is based on the patient's medical or dental history or if the clinical examination reveals cause for additional imaging.

Recall visit

Interproximal caries

High risk or clinically evident caries

- Children and Adolescents: 6-12 months.
- Adults: 6-18 months Low risk
- Children: 12-24 months
- Adolescents: 18-36 months.
- Adults: 24-36 months

Periodontal disease

- it is appropriate to make intraoral images, generally a combination of peri-apical and bite-wing images, to help establish and monitor the severity of the periodontal bone loss.

Special considerations

- Pregnancy

- Radiation Therapy

Refernce

- Text Book of Oral Radiology; White and Pharoah, Page no: 810-830

Case Scenario-1

- The first visit of a 5-year-old boy to a dental office. The patient's parent reports no known medical conditions, and the patient is cooperative. Clinical examination of the child reveals that there are open contacts between all the primary teeth and there is no evidence of caries. Because the child has grown up in a community that has a fluoridated water supply, a reasonably good diet is being observed, and the parent seems well motivated to promote good oral hygiene.

- no radiologic examination is required at this time. Images for the detection of developmental abnormalities are not in order at this age because a complete appraisal cannot be made at age 5 years. Even if it could be made, it may be too early to initiate treatment for such abnormalities.

Case Scenario-2

- A 25-year-old woman receiving a 6-month checkup after her last treatment for a fractured incisor. The patient does not have high-risk factors for caries, there are no clinical signs of caries, and no caries is evident on bitewing images made 6 months ago. In addition, the patient shows no evidence exists of periodontal disease or other remarkable signs or symptoms in general or associated with the recently fractured tooth.

- As long as the fractured incisor shows normal vitality testing, no radiographs are recommended for this patient. If the pulp of the tooth becomes nonvital, a periapical view of this tooth should be made.